MIL-M-23185C(OS)
7 May 1984
SUPERSEDING
MIL-M-23185B(OS)
14 September 1973
with AMENDMENT 2
16 November 1976

MILITARY SPECIFICATION

MARKER, LOCATION, SUBMARINE MK 21 MOD 0 MK 22 MOD 0, MK 23 MOD 0, AND MK 24 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This specification covers the Marker, Location, Submarine Mk 21 Mod 0, Mk 22 Mod 0, Mk 23 Mod 0 and Mk 24 Mod 0.

1.2 Definitions. The meaning of terms used in this specification shall be listed in 6.3.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Ammunition Lot Numbering
DOD-STD-2101	Classification of Characteristics

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein.

NAVAL ORDNANCE SYSTEMS COMMAND

DRAWINGS

DL 1331977	Marker, Location, Submarine, Mk 23 Mod 0 (Green)
DL 1331978	Marker, Location, Submarine, Md 22 Mod 0 (Yellow)
DL 13311979	Marker, Location, Submarine, Md 21 Mod 0 (Red)
DL 13311980	Marker, Location, Submarine, Md 24 Mod 0 (Black)
1332212	Pack, Shipping and Storage (for Fire Marker, Location, Submarine Mk 21, 22, 23, 24 or 27)
SA 2522933	Vibration Fixture

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained by the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

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3.1 Inspection and manufacture. The marker shall be in accordance with DL 1331977, DL 1331978, DL 1331979 or DL 1331980, as appropriate for the Mk and Mod specified (see 6.2.1), and this specification.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.3 Performance and product characteristics.

3.3.1 Performance.

3.3.1.1 Sealing. The marker shall withstand a vacuum of 6.0± 1.0 "Hg for a minimum period of 30 seconds (secs) without leakage when tested in accordance with 4.5.1.1.

3.3.1.2 Pressure-static functioning. The marker shall withstand a hydrostatic pressure of 550 psig minimum for a minimum period of 30 secs and within 4 hours thereafter meet the following requirements when tested in accordance with 4.5.1.2:

a. Pass through a tube 3.06" inside diameter, $36.00\ ^{+1.00}_{-0.00}$" long and straight within .003" (except guide shoe and latch assembly exterior).

b. Upon tripping the latch assembly, the valve shall pop out.

c. After tripping the latch assembly, the guide shoe shall rotate back of its own weight without binding.

d. Function (produce a smoke display).

e. Displays shall be of the color indicated by the stenciling and Mark of the marker.

f. The marker shall meet the delay and burning time requirements of TABLE I.

TABLE I. Delay and burning times.

Mark and Mod	Delay time (seconds) Maximum	Burning time (seconds) Minimum	Maximum
Mk 21 Mod 0	25	15	45
Mk 22 Mod 0	25	15	45
Mk 23 Mod 0	25	15	45
Mk 24 Mod 0	25	15	55

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3.3.2 Durability and environmental.

3.3.2.1 Forty (40) foot drop. The marker shall withstand a forty foot drop without burning, exploding or being unsafe to dispose of, when tested in accordance with 4.5.2.1.

3.3.2.2 Transportation vibration. The marker shall withstand transportation vibration without burning or exploding when tested in accordance with 4.5.2.2.

3.3.2.3 Temperature and humidity. The marker shall withstand temperature and humidity without burning or exploding when tested in accordance with 4.5.2.3. After the temperature and humidity test the markers shall meet the requirement of 3.3.1.2 except maximum burn time shall not exceed 70 secs.

3.4 Workmanship. The product, in all stages of production, shall be processed under such inspection control as to assure uniformity in quality. Parts shall be free of chips, dirts, grit or other foreign material. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents employed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection requirements. The inspection requirements are specified by the classified characteristics in the specification and on the referenced drawings, and also be the supplemental quality assurance provisions of the contract.

4.1.2 Inspection review. Unless otherwise specified in the contract, the contractor and all test activities identified in the contract shall develop a description of the proposed procedure and equipment to be used to perform all examinations and tests for review by the Technical Activity and Design Agent (see 6.2).

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article sample (see 4.3),

b. Quality conformance inspections (see 4.4).

4.3 First article sample. A first article sample consisting of 70 markers of one type designation shall be subjected to the examinations and tests specified in TABLE II.

4.4 Quality conformance inspection. An inspection lot shall consist of 26 to 3200 markers of one type designation, offered for delivery at one time including test samples (see 6.2). Inspection lots shall be inspected as follows:

a. Any change in materials, suppliers or production methods shall be considered sufficient interruption for the beginning of a new inspection lot. Record the reason for interruption on the Data Card (see 6.2).

b. Each lot shall be sampled as required for Plan A, TABLE II.

c. Upon rejection of any lot for test requirements, the next lot shall be tested and accepted in accordance with Plan A, TABLE II.

d. No lot shall be accepted in accordance with Plan B, TABLE II unless the preceding lot has met the test requirements of the applicable plan.

e. Each inspection lot shall contain squibs from one lot of manufacture.

f. Each inspection lot shall contain batteries from one lot of manufacture.

g. Sampling, examination, testing and acceptance of inspection lots shall be performed as specified in the steps given below:

STEP 1. Upon starting production or delivery of an acceptable first article sample, as applicable, all inspection lots shall be examined, tested in sequence and accepted in conformance with Plan A, TABLE II, until two (2) consecutive lots have been accepted.

STEP 2. Inspection lots other than those defined in Step 1 shall be examined, tested in sequence and accepted in conformance with TABLE II, Plan B, except that one lot shall be randomly selected from each five consecutive inspection lots and examined, tested in sequence, and accepted in conformance with TABLE II, Plan A.

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TABLE II. Inspection plans.

Examinations & tests	First article sampling plans		Inspection lot sampling plans			
Test sequence 6/	Sample size	Acceptance Criteria 1/3/	Plan 2/ A	Plan 2/ B	Sample size	Acceptance Critical 1/3/
Examinations and Workmanship (3.4) (4.6)	70 Markers	Critical Ac 0 Re 1 Major Ac 1 Re 2 Minor Ac 3 Re 4	X	X	5/	5/
Sealing test (4.5.1.1)		Ac 0 Re 1			100% of Inspection lot sample	Ac 0 Re 1
Forty (40) foot drop test (4.5.2.1)	5 Markers	Ac 0 Re 1	X		5 Markers	Ac 0 Re 1
Transportation Vibration test (4.5.2.2) Pressure-static Functioning test (4.5.1.2)	20 Markers	Ac 0 Re 1 (a&e) Ac 0 Re 1 (d) Ac 1 Re 2 (b, c&f) Ac 2 Re 3	X		Mil-STD-105 Level S-3	Ac 0 Re 1 (a&e) Ac 0 Re 1 (d) AQL 2.5 (b, c&f) AQL 4.0
Temperature and humidity test (4.5.2.2) *Pressure-static* Functioning test (4.5.1.2)	20 Markers	Ac 0 Re 1 (a&e) Ac 0 Re 1 (d) Ac 1 Re 2 (b, c&f) Ac 2 Re 3	X		MIL-STD-105 Level S-3	Ac 0 Re 1 (a&e) Ac 0 Re 1 (d) AQL 2.5 (b, c, &f) AQL 4.0
Pressure-Static functioning test (4.5.1.2)	20 Markers	(a&e) Ac 0 Re 1 (d) Ac 1 Re 2 (b, c&f) Ac 2 Re 3	X	X	MIL-STD-105 Level S-4	(a&e) Ac 0 Re 1 (d) AQL 2.5 (b, c&f) AQL 4.0
Grouped data acceptance 4/ (4.5.1.2)	60 Markers	(d) Ac 2 Re 3 (b,c&f) Ac 5 Re 6			None	None

NOTES:

1/ Ac = Acceptance number of defectives of specified requirements of Section 3.
Re = Rejection number for defectives of specified requirements of Section 3.
AQL = Acceptance quality levels are specified as percent defective.

2/ Test plans applicable to inspection lots.

3/ Acceptance criteria shall apply to characteristics (see 3.3.1.2 subparagraphs) enclosed in parentheses as a group.

4/ Grouped for pressure-static functioning test.

5/ Quality assurance inspection requirements of the contract apply.

6/ Test to be performed in order of indentured sequence within individual blocks.

See paragraph 3.3.1.2 for letter designations a through f.

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4.5 Classification of characteristics. Characteristics verified by inspections and tests specified herein are classified as Critical, Major and Minor in accordance with DOD-STD-2101. Critical characteristics are identified by the symbol (C), and Major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Charactertics that are not annotated by the classification code symbol are classified as Minor.

4.5.1 Performance tests.

4.5.1.1 Sealing test (C1). The marker shall be tested in a tank of distilled water which is changed daily and is equipped for observation under vacuum.

CAUTION: The water tank shall be equipped so that the marker must be raised above the water level while still under vacuum before releasing the vacuum and removing the marker.

The marker shall be positioned so that all surfaces shall be submerged 2 to 6" below the water surface and subjected to a vacuum of 6.0 ± 1.0" of Hg for a minimum period of 30 secs. Leakers are indicated by a constant stream of air bubbles issuing from the marker. Do not mistake the escape of occluded air for leakage. Defectives are markers failing to meet the requirements of 3.3.1.1.

4.5.1.2 Pressure-static functioning test. The marker shall be completely submerged in tap water at ambient temperature and subjected to a hydrostatic gage pressure of 550 psi minimum for a minimum period of 30 secs.

CAUTION: Pressure testing shall be conducted with the equipment in a test cell and performed remotely to protect personnel from explosive hazards.

The marker shall then be gaged for dimensional requirements as specified. Within four hours after pressure test, the marker shall be functioned by removing the two cotter pins and holding the marker in a vertical position, nose end up. Trip the latch assembly and move the marker to a horizontal position with the valve opening up. Verify that the valve pops out and the guide shoe falls back to a position parallel or below the periphery of the marker. (Shunt plunger shall move from the shunted position.) Personnel shall remain at a safe distance of 30' minimum when the marker is dropped into a container of salt water a minimum of 36" deep (see 6.3). The delay time shall be measured from dropping the marker into salt water until the nose plug blows off. For each marker tested, record failing to gage (C2), valve pop-out (M101), guide shoe rotation (M102), and function (M103); also record the delay time, and burning time (M104), and color display (C3). Defectives are markers failing to meet the requirements of 3.3.1.2.

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4.5.2 Durability and enviornmental tests.

4.5.2.1 Forty foot drop test (C4). The marker shall be subjected to the forty foot drop test prescribed in Test 103 of MIL-STD-331, except the marker shall be dropped free-fall without guidance system or associated equipment. Defectives are markers failing to meet the requirments of 3.3.2.1.

4.5.2.2 Transporation vibration test (C5). The marker, using test fixture SA 2522933 shall be subjected to the transportation vibration test prescribed in Test 401, Procedure II of MIL-STD-331 vertical position, nose down. Defectives are markers failing to meet the requirements of 3.3.2.2.

4.5.2.3 Temperature and humidity test (C6). The marker shall be subjected to one 14-day temperature and humidity cycle prescribed in Test 105 of MIL-STD-331 except that the Mk 22 marker shall be subjected to a temperature range of -40° to 140°F throughout the test cycle. Defectives are markers failing the meet the requirements of 3.3.2.3.

4.6 Examinations.

4.6.1 Marker examinations. Examinations shall be made to ascertain that the markers meet the requirements of drawings listed on DL 1331977, DL 1331978, DL 1331979, or DL 1331980, as applicable. Only markers with no defects shall be used for further testing or returned to the lot.

4.6.2 Packing and marking examinations. The packing of the marker and marking of the pack shall conform to Section 5 of this specification.

5. PACKAGING

5.1 Packing. Five markers shall be packed in accordance with dwg 1332212.

5.2 Ammunition data card. Data cards shall be prepared in accordance with MIL-STD-1167 (see 6.2).

5.3 Ammunition lot number. Ammunition lot number shall be in accordance with MIL-STD-1168.

6. NOTES

6.1 Intended use. The marker is intended for use by submarines to indicate submerged position to surface or air units. It consists of an aluminum body containing a pyrotechnic element which produces a smoke display of the designated color when hydrostatically and electrically initiated by two seawater-actuated batteries.

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6.2 Ordering data. Procurement document should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of specification.

b. Mark and mod required. (see 3.1 and 2.1.2).

c. Quantity of markers being ordered.

d. First article sample, if applicable, (see 3.2)

e. Test facility where samples are to be shipped for inspection (see 4.1)

f. Weapons Quality Engineering Center coordinating surveillance.

g. Inspection review, if applicable (see 4.1.2)

h. That the safety precaution requirements of the Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by the government activity, the safety precaution requirements of Ammunition and Explosives Ashore, OP 5, should be made applicable.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.	Option
4.3, 4.4	Reports, Test	DI-T-2072	--
5.2	Ammunition Data Card	DI-L-1410	--

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connections with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of 70 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the documents first article.

6.3 Definitions.

6.3.1 Marker. The term "marker" means Marker, Location, Submarine Mk 21 Mod 0, DL 1331979; Mk 22 Mod 0, DL 1331978; Mk 23 Mod 0, DL 1331977; or Mk 24 Mod 0, DL 1331980, as applicable.

6.3.2 Salt water. The term "salt water" means a salt water solution prepared by the following formula:

Ingredients	Precent by weight
Sodium chloride, 0-5-1926, Type II	3.2 ± 0.2
Water	96.8 ± 0.2

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing Activity:
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Project Number:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-23185C(OS)	Marker Location, Submarine Mk 21 Mod 0, Mk 22 Mod 0, Mk 23 Mod 0, and Mk 24 Mod 0

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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